FNCE30007 Derivative Securities
Tutorial 7 Solutions: BSM Model

QUESTION 1
Let X be a normally distributed random variable with mean 100 and variance 100.
(a) What is the probability that X will take on a value less than 85.
(b) What is the probability that X will take on a value greater than 90.
(c) What is the probability that X will take on a value less than 85 or more than 90.

A:

$X \sim \phi(100,100)$

$Z = \frac{x-u}{\sigma}$ then $Z \sim \phi(0,1)$

a) $Z = \frac{x-u}{\sigma} = \frac{85-100}{10} = -1.5$
$\mathrm{p}(\mathrm{x} < 85) = \mathrm{p}(\mathrm{z} < -1.5) = 0.0668$
b) $\mathrm{p}(\mathrm{x} > 90) = \mathrm{p}(\mathrm{z} > -1) = 1 - \mathrm{p}(\mathrm{z} < -1) = 1 - 0.1587 = 0.8413$
c) $p(x < 85) or\ p(x > 90) = 0.0668 + 0.8413 = 0.9081$

QUESTION 2
Let Z be a standard normal random variable.
(a) What is the probability that Z will take on a value between -1 and 1.
(b) What is the probability that Z will take on a value between -2 and 2.
(c) Comment on your answers in parts (a) and (b).

A:

a) $p(-1 < z < 1) = p(z < 1) - p(z < -1) = 0.8413 - 0.1587 = 0.6826$
b) $p(-2 < z < 2) = p(z < 2) - p(z < -2) = 0.9772 - 0.0228 = 0.9544$
c) The probabilities are as expected since
- Z is a normal random variable
- Z has a standard deviation of 1
- The probability of a normal random variable taking on a value within 1 (2) standard deviation of its mean is approximately 68% (95%)

QUESTION 3
The current price of a stock is $50. Assume the expected return on the stock is 18 % p.a. and its volatility is 30% p.a.. In the Black-Scholes framework, what is the probability distribution of the stock price in:
(a) one year time?
(b) two years' time?

A:

$\ln S_T \sim \phi\left[\ln S_0 + \left(\mu - \frac{\sigma^2}{2}\right)T, \sigma^2 T\right]$

a) When T=1,

Mean of $\ln S_T = ln50 + \left(0.18 - \frac{0.3^2}{2}\right) * 1 = 4.047$

Std dev $= 0.3\sqrt{1} = 0.3$

b) When T=2,

Mean of $\ln S_T = ln50 + \left(0.18 - \frac{0.3^2}{2}\right) * 2 = 4.182$

Std dev $= 0.3\sqrt{2} = 0.4243$

QUESTION 4

Consider a non-dividend paying stock when the stock price is $10.00, the risk-free interest rate is 6% p.a. and the volatility is 40% p.a..

a) use the Black-Scholes call option formula to calculate the current price of a 2-year European call option (strike $10.00) on the stock

b) use the Black-Scholes put option formula to calculate the current price of a 2 year European put option (strike $10.00) on the stock

c) use the simulator to check your answers.

A:

S=10 K=10 r=0.06 σ=0.4 T=2

a) $d_1 = \frac{\ln(10/10)+(0.06+0.4^2/2)*2}{0.4\sqrt{2}} = 0.49497475$

$d_2 = 0.49497475 - 0.4\sqrt{2} = -0.07$

$N(d_1) = 0.6879$

$N(d_2) = 0.4721$

$c = 10 * 0.6879 - 10\, e^{-0.06*2} * 0.4721 = 2.69$

b) $N(-d_1) = 0.3121$

$N(-d_2) = 0.5279$

$p = 10e^{-0.06*2} * 0.5279 - 10 * 0.3121 = 1.56$

Alternatively, use put-call parity

$p = c - S + Ke^{-rT}$

$p = 2.69 - 10 + 10e^{-0.06*2} = 1.56$

QUESTION 5

Reconsider the European call option in question 5(a). Recalculate the price of the option using:

a) a one-period binomial model.

b) a ten-period binomial model using the simulator

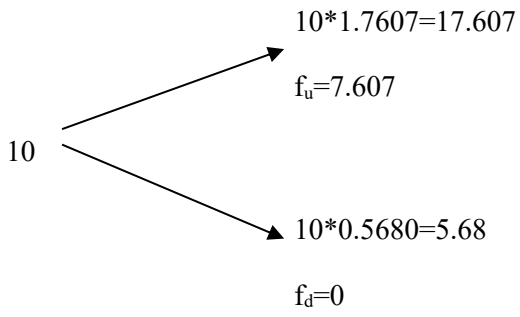
c) what do your answers in parts (a) and (b) suggest about the relationship between the binomial and Black-Scholes models?

A:

a) 1 period binomial model

$u = e^{\sigma\sqrt{\Delta t}} = e^{0.4\sqrt{2}} = 1.7607$

$d = \frac{1}{u} = 0.5680$



$p = \frac{e^{r\Delta t} - d}{u - d} = \frac{e^{0.06*2} - 0.5680}{1.7607 - 0.5680} = 0.4691$

$c = \frac{0.4691*7.607+0}{e^{0.06*2}} = 3.16$

b) 10-period binomial model

$\Delta t = \frac{T}{10} = 0.2$

$u = e^{\sigma\sqrt{\Delta t}} = e^{0.4\sqrt{0.2}} = 1.195884$

$d = \frac{1}{u} = 0.836202$

Using the simulator, you will get c=2.66

c) Parts a), b) and c) illustrate that increasing the time step in the binomial model leads to a convergence between the binomial option price and the BSM option price

QUESTION 6

Consider an option on a dividend-paying stock when the stock price is $30, the exercise price is $29, the risk-free interest rate is 5% per annum, the volatility is 25% per annum, and the time to maturity is four months. The stock is due to go ex-dividend in 1.5 months and the expected dividend is 50 cents.

a. **What is the price of the option if it is a European call? Check your answers using the simulator**
b. **What is the price of the option if it is a European put? Check your answers using the simulator**
c. **What happens to the price if the stock is expected to pay another 50 cents dividend in 6 months (ex-dividend date in 6 months)? Check your answers using the simulator.**

Comment on the findings.

a. The present value of the dividend must be subtracted from the stock price. This gives a new stock price of:

$$30 - 0.5e^{-0.125\times0.05} = 29.5031$$

and

$$d_1 = \frac{ln(29.5031/29)+(0.05+0.25^2/2)\times0.3333}{0.25\sqrt{0.3333}} = 0.3068 \approx 0.31$$

$$d_2 = 0.3068 - 0.25\sqrt{0.3333} = 0.1625 \approx 0.16$$

$$N(d_1) = 0.6217;\ N(d_2) = 0.5636$$

The price of the option is therefore

$$29.5031 * 0.6217 - 29 * e^{-0.05*0.3333} * 0.5636 = 2.2678$$

b. Because

$$N(-d_1) = 0.3783;\ N(-d_2) = 0.4364$$

the value of the option when it is a European put is

$$29 * e^{-0.05*0.3333} * 0.4364 - 29.5031 * 0.3783 = 1.2854$$

c. Your answer would stay the same as the next dividend will be paid after the option is expired.

QUESTION 7

Suppose that the spot price of the Canadian dollar is U.S. $0.95 and that the Canadian dollar/U.S. dollar exchange rate has a volatility of 8% per annum. The risk-free rates of interest in Canada and the United States are 4% and 5% per annum, respectively. Calculate the value of a European call option to buy one Canadian dollar for U.S. $0.95 in nine months. Use put-call parity to calculate the price of a European put option to sell one Canadian dollar for U.S. $0.95 in nine months. What is the price of a call option to buy U.S. $0.95 with one Canadian dollar in nine months? Using the simulator to check your answers.

In this case $S_0 = 0.95$, $K = 0.95$, $r = 0.05$, $r_f = 0.04$, $\sigma = 0.08$ and $T = 0.75$.

$$d_1 = \frac{ln(0.95/0.95) + (0.05 - 0.04 + 0.0064/2) \times 0.75}{0.08\sqrt{0.75}} = 0.1429 \approx 0.14$$

$$d_2 = d_1 - 0.08\sqrt{0.75} = 0.0736 \approx 0.07$$

and

$$N(d_1) = 0.5557, \quad N(d_2) = 0.5279$$

The value of the call, c, is given by

$c = 0.95e^{-0.04*0.75} * 0.5557 - 0.95 * e^{-0.05*0.75} * 0.5279 = 0.0293$

i.e., it is 2.93 cents. From put–call parity

$$p + S_0 e^{-r_f T} = c + Ke^{-rT}$$

so that

$$p = 0.0293 + 0.95 * e^{-0.05*0.75} - 0.95e^{-0.04*0.75} = 0.0224$$

The option to buy US$0.95 with C$1.00 is the same as an option to sell one Canadian dollar for US$0.95. This means that it is a put option on the Canadian dollar and its price is US$0.0224.

QUESTION 8

After reading the Virgin case, complete the following questions before the tutorial.

The Effect of Increasing the Number of Periods

Now focus on Option C: Jan 2027 Brent Call, which has the longest time to maturity.

Revalue the option using: N = 2, 4, 8, 16

8.1 For each value of N, recalculate Δt = T/N and the corresponding u, d and p. Use the simulator to construct the larger binomial trees where appropriate and compare the estimates in this table.

Number of Periods, N	**Binomial Value**	**Observed Market Price**	**Difference: Model - Market**	**Absolute % Pricing Error**
2		7.72		
4		7.72		
8		7.72		
16		7.72		

8.2 Interpretation

- How does the estimated option value change as the number of periods increases?
- Does the binomial estimate appear to become more stable as the tree becomes finer?
- How does the pricing error change as N increases?
- What does the pattern suggest about convergence of the binomial estimate as the tree is refined?

Data note: Option-chain figures are based on Barchart market data observed on 2 September 2026. Prices are quoted in USD per barrel. Barchart reports an option point value of USD 1,000.

Answer:

8.1. Reference Results

Number of Periods, N	Binomial Value	Observed Market Price	Difference: Model - Market	Absolute % Pricing Error
2	6.82	7.72	-0.90	11.61%
4	7.23	7.72	-0.49	6.31%
8	7.45	7.72	-0.27	3.45%
16	7.57	7.72	-0.15	1.99%

8.2 Suggested interpretation

As the number of periods increases, the estimated option value increases and becomes progressively more stable. The pricing error also falls across the values of N shown above.

Overall, the results illustrate convergence of the binomial estimate as the tree is refined.

Conclusion

As the number of periods increases, the binomial estimate becomes more stable and converges as the tree is refined.